The H-join of complex unit gain graphs and their stability

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Some basics

Gain graphs

A gain graph is a triple $\tilde{G} = (G, \mathcal{G}, \varphi)$ where

- $G = (V, E)$ is an underlying (usually simple) graph,
- $\mathcal{G}$ is a group (often written multiplicatively), called the gain group,
- φ is a function that assigns to each oriented edge $e = uv$ a group element $\varphi(e) \in \mathcal{G}$, called its gain, such that

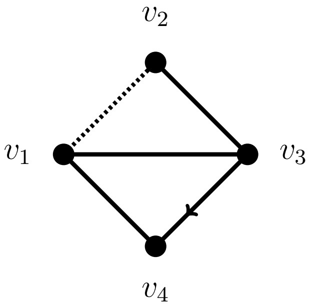
$$\varphi(uv) = \varphi(vu)^{-1}.$$

Here we will be considering the multiplicative group of complex units,

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}.$$

Adjacency matrices

The adjacency matrices for signed and gain graphs are defined in the natural way.



$$A_{\tilde{G}} = \begin{pmatrix} 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & i \\ 1 & 0 & -i & 0 \end{pmatrix}$$

When the gain function φ labels an edge between v_i and v_j with a group element, we have $\varphi(v_i v_j)$ in the ij^{th} position and $\varphi(v_j v_i) = \varphi(v_i v_j)^{-1}$ in the ji^{th}. We can see how this works between v_3 and v_4 here.

Gain balance and switching

Two gain graphs are switching equivalent if and only if their adjacency matrices are conjugated by a diagonal matrix with diagonal entries in the group. This is analogous to the switching for signed graphs, where the entries were only $+1$ and -1.

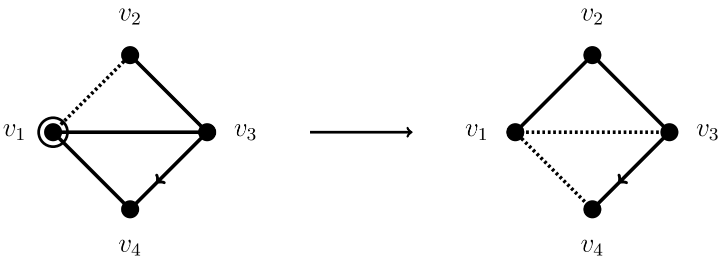
In other terms, two $\mathbb{T}$-gain graphs $\tilde{G}$ and $\tilde{G}'$ on the same underlying graph are switching equivalent if there exists a function $s : V_G \to \mathbb{T}$ such that for any pair of adjacent vertices v_i and v_j we have

$$\varphi'(v_i v_j) = s(v_i)^* \varphi(v_i v_j) s(v_j).$$

A gain graph is said to be balanced if the product of edge signs in every closed walk is equal to the identity element of the group. Again this resembles what we have for signed graphs, where the product of edge signs in cycles must be equal to $+1$ for balance.

Gain switching example

Here's an example, performing switching on v_1 by multiplying by -1.



Closed walks

In a graph G, a closed walk W of length k is a list of vertices $v_1v_2\dots v_{k+1}$ such that $v_{k+1} = v_1$ and $v_i \sim v_{i+1}$ for each $i = 1, \dots, k$.

If G is endowed with a complex unit gain function φ, it is possible to define the gain of a closed walk as $\varphi(W) = \varphi(v_1v_2)\varphi(v_2v_3)\dots\varphi(v_kv_{k+1})$.

Given in terms of the gains of closed walks, two gain functions φ and φ' are switching equivalent if and only if $\varphi(W) = \varphi'(W)$ for every closed walk W in G.

A gain graph is balanced if and only if the product of the edge gains in every closed walk is equal to the identity element of the gain group.

Equivalently, a complex unit gain graph with gain function φ is balanced if and only if it can be switched to the all-positive underlying graph.

The H-join

Nice join operations

Typically, a well-designed generalisation of a product or join operation on gain graphs should aim to satisfy as many of the following three properties as possible.

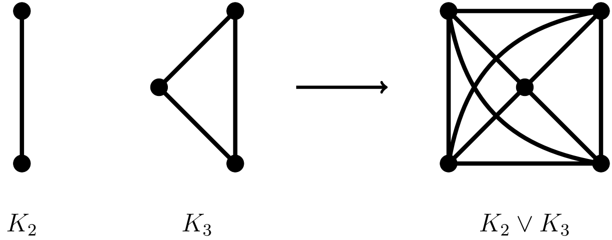
- If the gain functions on the component graphs are all equal to the identity function, the gain function on the product assigns the identity element to all of its edges.
- Geometric and spectral properties are preserved or suitably generalised.
- The operation is switching-stable, meaning that replacing the gain functions of the components with switching equivalent ones does not change the switching equivalence class of the product.

This last property is particularly useful in spectral computations since it is well known that switching equivalent gain graphs are cospectral.

Standard join

Usually, considering two graphs G_1 and G_2, the join $G_1 \vee G_2$ is taken to be the union of the existing vertices and edges along with new edges connecting each vertex of G_1 to every vertex of G_2.

Here we have K_2, K_3, and the join $K_2 \vee K_3$.



H-join

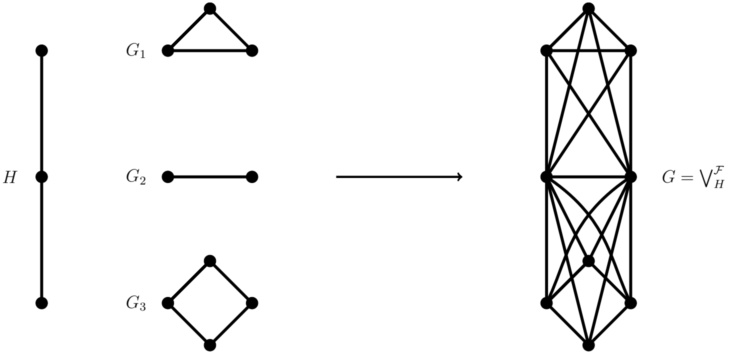
A generalisation of the standard join operation works in the following way.

- Consider a graph H of order k and a family of graphs $\mathcal{F} = \{G_1, \dots, G_k\}$.
- Replace each vertex v_i of H with the graph G_i.
- Connect each vertex of G_i to every vertex of G_j when v_i and v_j are adjacent in H.

Hence, when $H = K_2$, the H-join corresponds precisely to the usual join graph operation.

H-join

Here we have the H-join of $\mathcal{F} = \{K_3, K_2, C_4\}$ with $H = P_3$.



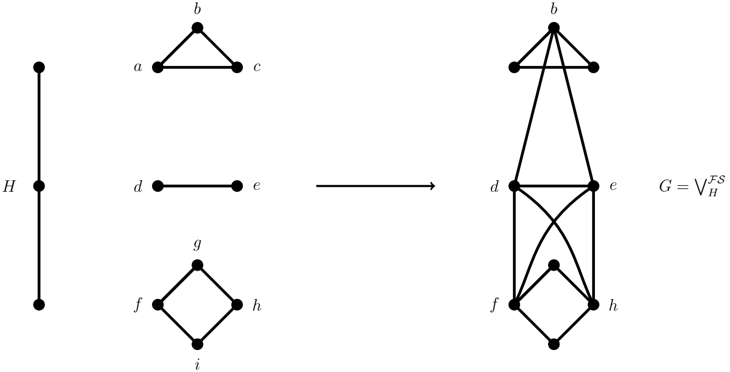
H-generalised join

The H-generalised join was introduced in 2013 by Cardoso et al. The main idea is similar, but this time rather than making joins between all the vertices of the component graphs, only a select subset of their vertices are used.

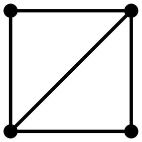
- Consider a graph H of order k and a family of graphs $\mathcal{F} = \{G_1, \ldots, G_k\}$.
- Consider a family of vertex subsets $\mathcal{S} = \{S_1, \ldots, S_k\}$, such that $S_i \subseteq V_{G_i}$.
- We substitute in each graph G_i for each vertex v_i of H and then make the joins between the vertices in the S_is where the edges exist in H.

H-generalised join

Here we have the H-generalised join of $\mathcal{F} = \{K_3, K_2, C_4\}$ with $H = P_3$ and $\mathcal{S} = \{\{b\}, \{d, e\}, \{f, h\}\}$.



Indexing maps



$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A further generalisation is known as the H_m-join, and in this situation the operation is constrained by indexing maps.

For the vertices of each G_i we have a map M_i, such that $M_i : V_{G_i} \to \{1, \ldots, m\}$. The m in the H_m-join refers to this largest value that a vertex can be sent to by the indexing maps.

The corresponding indexing matrices E_i are defined so that $(E)_{st} = 1$ if $M(v_s) = t$ and $(E)_{st} = 0$ otherwise. Hopefully this is a little easier to understand by looking at the diagram above.

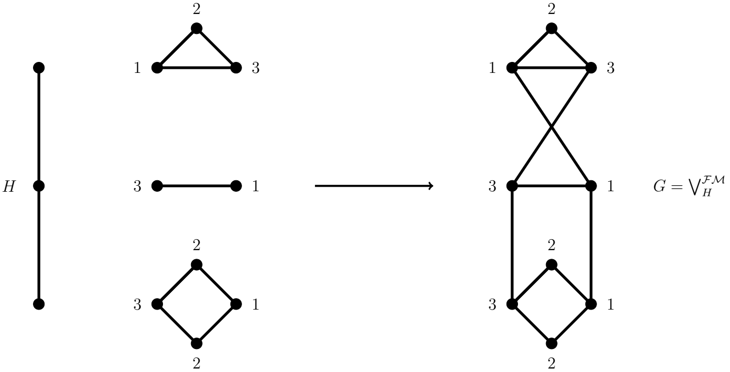
H_m-join

The H_m-join was introduced in 2024 by Arunkumar and Ganeshbabu. The idea is again similar, but this time rather than making joins between all the vertices within given subsets, each vertex has an index value, and then those with equal values that are in adjacent graphs are joined.

- Consider a graph H of order k and a family of graphs $\mathcal{F} = \{G_1, \ldots, G_k\}$.
- Consider a family of index maps $\mathcal{M} = \{M_1, \ldots, M_k\}$, such that $M_i : V_{G_i} \to \{1, \ldots, m\}$.
- We substitute in each graph G_i for each vertex v_i of H and then make the joins between the vertices in the G_is where the edges exist in H and the vertices have equal index values.

H_m-join

Here we have the H_m-join of $\mathcal{F} = \{K_3, K_2, C_4\}$ where $H = P_3$ with respect to the indexing maps M_1, M_2, M_3 whose values are labelled on the vertices.



$H \to H$-generalised $\to H_m$

We can see that the H-join operation is the H-generalised operation where $S_i = V_{G_i}$ in all cases, and is the H_m-join operation where $m = 1$ and $M : V_{G_i} \to \{1\}$.

Consider an H-generalised join with H as a graph of order k, and let $m = k + 1$. For each $i \in \{1, \ldots, k\}$ we can define $M_i(v) = 1$ if $v \in S_i$ and $M_i(v) = i + 1$ otherwise. Then we can see that the H-generalised operation is an H_m-join under these conditions.

Various graph families can be realised as H_m-joins. For instance generalised Petersen graphs, wheel graphs, lollipop graphs, tadpole graphs, Helm graphs, and so on.

H-join adjacency matrix

The adjacency matrix of the H-join is essentially obtained by replacing each entry of A_H with a block matrix. Along the diagonal we put the adjacency matrices of the G_i, where there is a 1 we substitute in an all-one matrix of the relevant size, and where there is a 0 we substitute in an all-zero matrix.

Let G be the H-join of $\mathcal{F} = \{K_3, K_2, C_4\}$ with $H = P_3$ from before. Then we have

$$A_H = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A_G = \left(\begin{array}{ccccccccc} 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ \hline 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ \hline 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \end{array}\right).$$

H-join history with ordinary graphs

1974: The first iteration of the operation we are studying was introduced as the generalised composition of graphs by Schwenk.

2011: Cardoso et al. found the adjacency spectra of the H-join of a family of regular graphs with H being restricted to a path graph.

2013: Cardoso et al. reintroduced the procedure and gave it the name of H-join. They also derived the spectra of H-joins of a family of regular graphs but this time with no restriction on H.

2013: A variation known as the H-generalised join was presented by Cardoso et al. Here the adjacencies between component graphs are restricted according to vertex subsets.

H-join history with ordinary graphs

2014: Both the signless Laplacian ($Q_G = D_G + A_G$) and the normalised Laplacian ($\mathcal{L}_G = D_G^{-1/2} L_G D_G^{-1/2}$) spectra of H-joins were determined by Wu et al.

2022: Cardoso et al. were able to characterise the universal adjacency ($U_G = \alpha A_G + \beta I + \gamma J + \delta D_G$, with $\alpha, \beta, \gamma, \delta \in \mathbb{R}$, and $\alpha \neq 0$) spectrum of a graph obtained by the H-join operation, now to a family of arbitrary graphs.

2024: More recently, Arunkumar and Ganeshbabu proposed a further generalisation of this operation, the H_m-join, this time constraining adjacencies according to indexing maps on the vertices of the component graphs.

H-join history with signed graphs

2019: Brunetti et al. computed both the adjacency and Laplacian spectra of signed graphs obtained from the lexicographic product operation, something similar to the H-join but more limited.

2021: For the $\dot{H}$-join proper on signed graphs, the operation was introduced and the adjacency and Laplacian spectra characterised by Zhang et al.

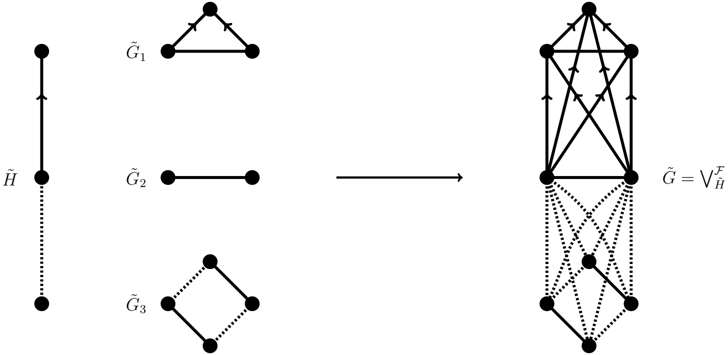
2023: Li et al. produced signed graphs by performing the $\dot{H}$- and $\dot{H}$-generalised join operations and were able to determine the characteristic polynomials and spectra of many related matrices.

2025: I have worked on the signed $\dot{H}_m$-join and the spectra of several of its matrices, this paper was published just a couple of months ago.

$\tilde{H}$-join of complex unit gain graphs

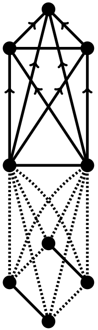
The gain variant of this operation, which we may denote by $\bigvee_{\tilde{H}}^{\mathcal{F}}$ or $\bigvee_{\tilde{H}}\{\tilde{G}_1, \ldots, \tilde{G}_k\}$, works in a similar way.

Only now we have to make the joins with respect the gain functions of the edges in $\tilde{H}$, here is an example using the same underlying graphs as before.



$\tilde{H}$-join adjacency matrix

The adjacency matrix works similarly too, only now we may now have blocks of any complex unit $z \in \mathbb{T}$, not just 1 and 0.



$$A_{\tilde{G}} = \left(\begin{array}{ccccccccc}
0 & i & 1 & -i & -i & 0 & 0 & 0 & 0 \\
-i & 0 & -i & -i & -i & 0 & 0 & 0 & 0 \\
1 & i & 0 & -i & -i & 0 & 0 & 0 & 0 \\
\hline
i & i & i & 0 & 1 & -1 & -1 & -1 & -1 \\
i & i & i & 1 & 0 & -1 & -1 & -1 & -1 \\
\hline
0 & 0 & 0 & -1 & -1 & 0 & -1 & 0 & 1 \\
0 & 0 & 0 & -1 & -1 & -1 & 0 & 1 & 0 \\
0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 & -1 \\
0 & 0 & 0 & -1 & -1 & 1 & 0 & -1 & 0
\end{array}\right).$$

Following our example, we have $\varphi_H(12) = -i$, $\varphi_H(21) = i$, $\varphi_H(23) = -1$, $\varphi_H(32) = -1$. Thus the adjacency matrix is as above.

$\tilde{H}$-join characteristic polynomial

Let $\tilde{G}$ be the $\tilde{H}$-join of a family $\mathcal{F} = \{\tilde{G}_1, \ldots, \tilde{G}_k\}$.

Define the characteristic polynomial of $\tilde{G}_i$ for each $i = 1, \ldots, k$ as $p_i = \det\left(\lambda I_{n_i} - A_{\tilde{G}_i}\right)$ and denote $\chi_i(\lambda) = \mathbf{1}_{n_i}^T(\lambda I_{n_i} - A_{\tilde{G}_i})^{-1}\mathbf{1}_{n_i}$.

Then, the characteristic polynomial of $\tilde{G}$ is

$$p_{\tilde{G}}(\lambda) = \left(\prod_{i=1}^{k} p_i\right)\det(M),$$

where $M = (m_{ij})$ is the $k \times k$ matrix with

$$m_{ij} = \begin{cases} 1, & \text{if } i = j, \\ -\varphi_H(v_i v_j)\chi_i, & \text{if } i \neq j. \end{cases}$$

$\tilde{H}$-join adjacency eigenvalues

Let M be an $n \times n$ normal complex matrix and let $\mathbf{u}$ be an $n \times 1$ complex vector.

An eigenvalue λ of M is called a **u**-main eigenvalue if the corresponding eigenspace $\xi_M(\lambda)$ is not orthogonal to $\mathbf{u}$. Otherwise we say it is **u**-non-main.

Consider $\tilde{G} = \bigvee_{\tilde{H}}\{\tilde{G}_1, \ldots, \tilde{G}_k\}$. Then

- if λ is a non-main eigenvalue of $\tilde{G}_i$ of multiplicity μ, it follows that λ is an eigenvalue of $\tilde{G}$ of multiplicity at least μ,
- if λ is a main eigenvalue of $\tilde{G}_i$ of multiplicity μ, it follows that λ is an eigenvalue of $\tilde{G}$ of multiplicity at least $\mu - 1$,
- the remaining eigenvalues of $\tilde{G}$ are precisely the roots of the polynomial $\det(\Omega(\lambda))$ (which I don't have enough time to detail here).

r-$\mathbb{T}$-regular graphs

Let u be a vertex of a $\mathbb{T}$-gain graph $\tilde{G} = (G, \mathbb{T}, \varphi)$. Brunetti defined the numbers

$$d^{\rightarrow}_{\tilde{G}}(u) = \sum_{uv \in E_G} \varphi(uv) \qquad \text{and} \qquad d^{\leftarrow}_{\tilde{G}}(u) = \sum_{uv \in E_G} \varphi(vu)$$

to be, respectively, called the $\mathbb{T}$-outgain and the $\mathbb{T}$-ingain of the vertex u.

Further to this, we say that an r-$\mathbb{T}$-regular graph to be a $\mathbb{T}$-gain graph $\tilde{G} = (G, \mathbb{T}, \varphi)$ such that $d^{\rightarrow}_{\tilde{G}}(u) = r$ for all $u \in V_G$.

Consider the $\tilde{H}$-join of a family of r_i-$\mathbb{T}$-regular graphs. Since $A_{\tilde{G}_i}$ has constant row sum r_i, the all-one vector $\mathbf{1}_{n_i}$ will be an eigenvector with eigenvalue r_i, and so r_i will clearly be a main eigenvalue of $\tilde{G}_i$.

In addition, since $A_{\tilde{G}_i}$ is Hermitian, all of the other eigenvectors will be orthogonal to $\mathbf{1}_{n_i}$, meaning all of the other eigenvalues of $\tilde{G}_i$ will be non-main.

r-$\mathbb{T}$-regular graphs

Thus, let $\tilde{G} = \bigvee_{\tilde{H}}\{\tilde{G}_1, \dots, \tilde{G}_k\}$ be the $\tilde{H}$-join of a family of vertex-disjoint graphs $\mathcal{F} = \{\tilde{G}_1, \dots, \tilde{G}_k\}$ in which $\tilde{G}_i = (G_i, \mathbb{T}, \varphi_i)$ is an r_i-$\mathbb{T}$-regular graph of order n_i for each $i = 1, \dots, k$.

We have

$$\mathrm{Spec}_{\tilde{G}} = \left(\bigcup_{i=1}^{k}\left(\mathrm{Spec}_{\tilde{G}_i} \setminus \{r_i\}\right)\right) \cup \mathrm{Spec}_{\Psi},$$

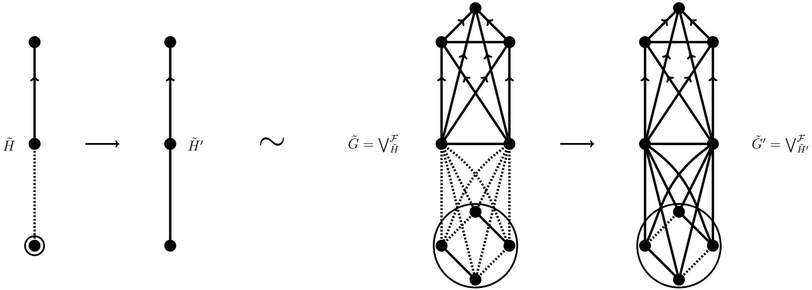
where $\Psi = (\psi_{ij})$ is the $k \times k$ matrix with

$$\psi_{ij} = \begin{cases} r_i, & \text{if } i = j, \\ \varphi_H(v_i v_j) n_i, & \text{if } i \neq j. \end{cases}$$

We can see that all but one (namely, r_i) of the eigenvalues for each component graph will be carried to the spectrum of the product, and we can directly compute those which remain.

Stability under switching of $\tilde{H}$

Let $\tilde{H} = (H, \varphi_H)$ and $\tilde{H}' = (H, \varphi'_H)$ be switching equivalent. Then $\tilde{G} = \bigvee^{\mathcal{F}}_{\tilde{H}}$ and $\tilde{G}' = \bigvee^{\mathcal{F}}_{\tilde{H}'}$ are also switching equivalent.



The proof is simple enough and the claim can be understood straightforwardly with an example.

A switching-stable $\tilde{H}$-join

Pseudo-potentials

Let $\tilde{G} = (G, \mathbb{T}, \varphi)$ be a connected $\mathbb{T}$-gain graph. Fix a spanning tree $\mathcal{T}$ of G and fix a vertex v in V_G.

The pseudo-potential $U_{\mathcal{T},v,\varphi} : V_G \longrightarrow \mathbb{T}$ is the function defined, for each w in V_G, as

$$U_{\mathcal{T},v,\varphi}(w) = \varphi(\gamma_{vw}),$$

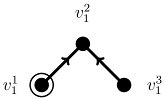
where γ_{vw} is any walk contained in $\mathcal{T}$ from v to w.

If $\tilde{G}$ is not connected we require a maximal spanning forest instead of a single spanning tree and a list of fixed vertices $\overline{v}$ (one for each connected component) instead of a single vertex.

Pseudo-potentials example

Keeping with our same example, here are some spanning trees with fixed vertices and the corresponding values of pseudo-potentials.

$\mathcal{T}_1$

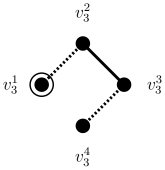


$$U_1(v_1^1) = 1,\ U_1(v_1^2) = i,\ U_1(v_1^3) = 1,$$

$\mathcal{T}_2$

$$U_2(v_2^1) = 1,\ U_2(v_2^2) = 1,$$

$\mathcal{T}_3$



$$U_3(v_3^1) = 1,\ U_3(v_3^2) = -1,$$
$$U_3(v_3^3) = -1,\ U_3(v_3^4) = 1.$$

Switching-stable definition

Now that we have the pseudo-potentials, this is how we construct our switching-stable version of the operation.

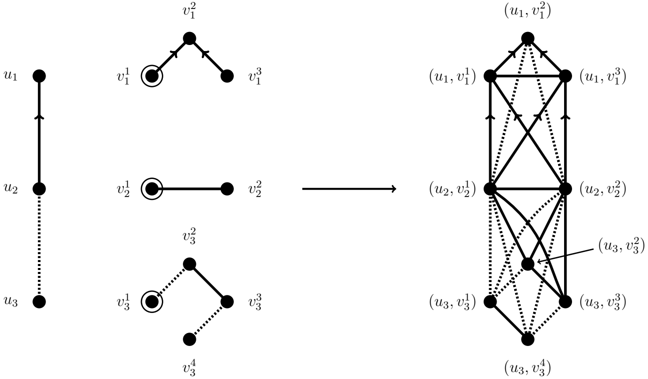
- Like before, consider a graph $\tilde{H}$ of order k and a family of graphs $\mathcal{F} = \{\tilde{G}_1, \ldots, \tilde{G}_k\}$.
- Fix a spanning tree $\mathcal{T}_i$ of each G_i, fix a vertex v_i^s in each V_{G_i}, and take the pseudo-potential functions $U_i = U_{\mathcal{T}_i, v_i^s, \varphi_i} : V_{G_i} \to \mathbb{T}$ as $U_i(w_i^t) = \varphi_i(\gamma_{v_i^s w_i^t})$ for each w_i^t in V_{G_i}.
- Replace each vertex v_i of $\tilde{H}$ with the graph $\tilde{G}_i$.
- Connect each vertex of $\tilde{G}_i$ to every vertex of $\tilde{G}_j$ when v_i and v_j are adjacent in $\tilde{H}$, with gain function φ satisfying

$$\varphi((u_i, v_i^s), (u_j, v_j^t)) = \begin{cases} \varphi_i(v_i^s v_i^t), & \text{if } i = j \text{ and } v_i^s v_i^t \in E_{G_i}, \\ \varphi_H(u_i u_j) U_i(v_i^s)^{-1} U_j(v_j^t), & \text{if } u_i u_j \in E_H. \end{cases}$$

Example

Here is the explicit computation for one of the edges

$$\varphi((u_2, v_2^1), (u_1, v_1^2)) = \varphi_H(u_2 u_1) U_2(v_2^1)^{-1} U_1(v_1^2) = i \cdot 1 \cdot i = -1.$$



Stability under switchings of $\tilde{G}_i$

The key thing here is that now we have an operation which is switching-stable with regards to not only $\tilde{H}$ but also to the component $\tilde{G}_i$ graphs.

One way to see this is to consider replacing each φ_i with a switching equivalent gain function φ_i'. Then there exist functions $s_i : V_{G_i} \to \mathbb{T}$, such that

$$\varphi_i'(uv) = s_i^*(u)\varphi_i(uv)s(v).$$

The pseudo-potentials, which are determined by the gain functions, are altered under switching by a multiplication of s^*, such that

$$U_i'(v) = s_i^*(v)U_i(v) = U_i(v)s_i^*(v).$$

Stability under switchings of $\tilde{G}_i$

Substituting these values with the original gain functions, we get $\varphi'((u_i, v_i^s), (u_j, v_j^t))$ for the entire H-join as

$$\begin{cases} s_i^*(v_i^s)\varphi_i(v_i^s v_i^t)s_i(v_i^t), & \text{if } i = j \text{ and } v_i^s v_i^t \in E_{G_i}, \\ s_i(v_i^s)U_i(v_i^s)^{-1}\varphi_H(u_i u_j)U_j(v_j^t)s_j^*(v_j^t), & \text{if } u_i u_j \in E_H. \end{cases}$$

Now considering all closed walks in the two graphs, their original gain functions and the new ones: since the switching factors and their complex conjugates appear in pairs for each vertex in the walk (once upon arrival and once again upon leaving) we have their gains remaining the same, and this shows precisely that the two graphs are switching equivalent.

Furthermore, since the operation is switching-stable, we can compute the spectrum without loss of generality by considering the graph obtained by replacing each $\tilde{G}_i$ with a $\mathcal{T}$-positive (nice) representative.

Comparing stability and balancedness

So, at the cost of fixing a spanning tree, we now have a method for constructing a nicer product.

	standard $\tilde{H}$-join	stable $\tilde{H}_{\mathcal{T},\overline{v}}$-join
stable	only under switchings of $\tilde{H}$	under switchings of $\tilde{H}$ and each component $\tilde{G}_i$
balanced	if and only if $\tilde{H}$ is balanced and each $\tilde{G}_i$ is all-positive	if and only if $\tilde{H}$ is balanced and each $\tilde{G}_i$ is balanced

Future ideas

Some potential avenues for further research include:

- establish results for different matrices, such as the Laplacian,
- extending this work to more general groups rather than just complex units,
- studying the H-generalised and H_m variants of this operation when working with gain graphs,
- applying the methods developed here to fix stability issues in other existing products of signed and gain graphs,
- creating new, nicely defined join operations of graphs and looking into those.

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Thank you!